Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/31/2006
PI Name:	Paik, Ho Jung Ph.D.		
Project Title:	Search for Spin-Mass Interaction With A Superconducting Differential Angular Accelerometer		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Gravitational & relativistic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hpaik@physics.umd.edu	**Fax:**	FY 301-405-6087
PI Organization Type:	UNIVERSITY	**Phone:**	301-405-6086
Organization Name:	University of Maryland		
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PI Address 2:			
PI Web Page:			
City:	College Park	**State:**	MD
Zip Code:	20742	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	96-HEDS-03
Start Date:	02/01/1998	**End Date:**	12/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JPL
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Paik, Ho Jung		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the fromtiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
Task Description:	In order to search for the spin-mass interaction, the intrinsic spin of electrons in a magnetic material will be modulated and its effect on a nearby test mass will be monitored. The spin source will be made out of soft ferromagnets. A very sensitive superconducting differential angular accelerometer will be constructed as the force detector. According to Einstein's theory of general relativity, mass interacts only with mass through gravity. However, some modern theories of elementary particles predict the existence of a new particle, such as the "axion," which mediates interaction between mass and intrinsic spin of a particle. The objective of this project is the development of an instrument capable of detecting spin-mass interaction at the level predicted for axions. Detection of spin-mass interaction will test limits of general relativity and provide a crucial test for these new theories of particles.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 10/30/2007)